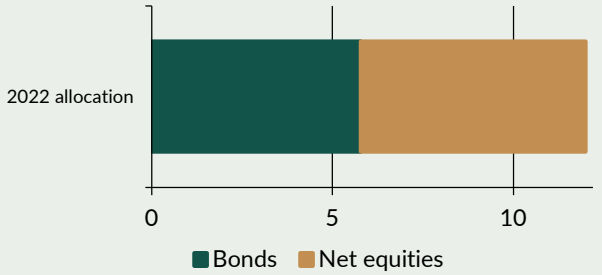
KAPPA FOUNDATION

Investment recommendation and revised mandate

Where the $12m starts



$5.784m bonds

$6.210m equities

$6,210 entry fee

The $12 million allocation that puts our payment dates first.

Lecturer: Dr. Omer Cayirli
Research & Teaching Assistant: Ty Dam Phuong Ngoc

Our recommendation

Meet the obligations without changing the frozen stock weights.

SELECTED POLICY

Principal matching

2,400 Beta 1,200 Gamma 2,400 Delta

$6.21m

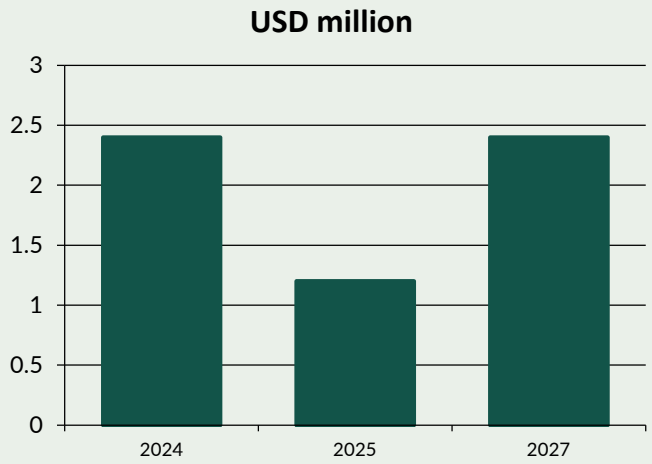
Net equity investment

4.80%

Required equity CAGR to 2030

A $396k cash reserve remains after the 2025 commitment.

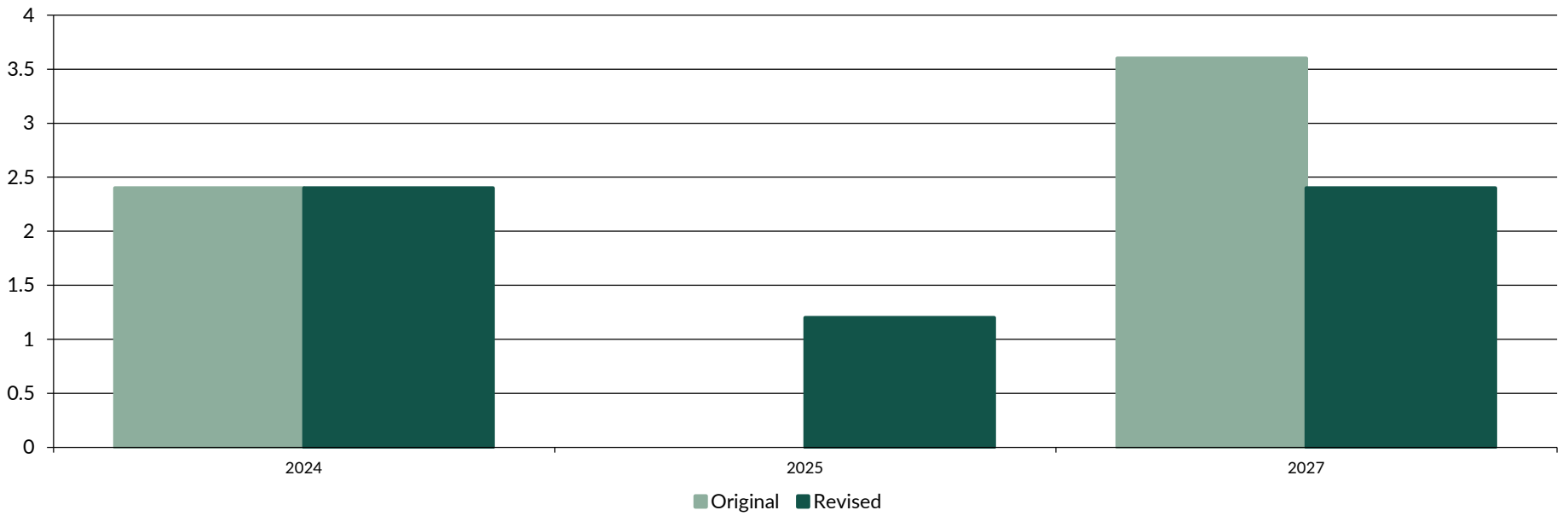
Revised commitments



$6.0m across three dates

The $6m total stays the same

Only the timing of the second commitment has changed.

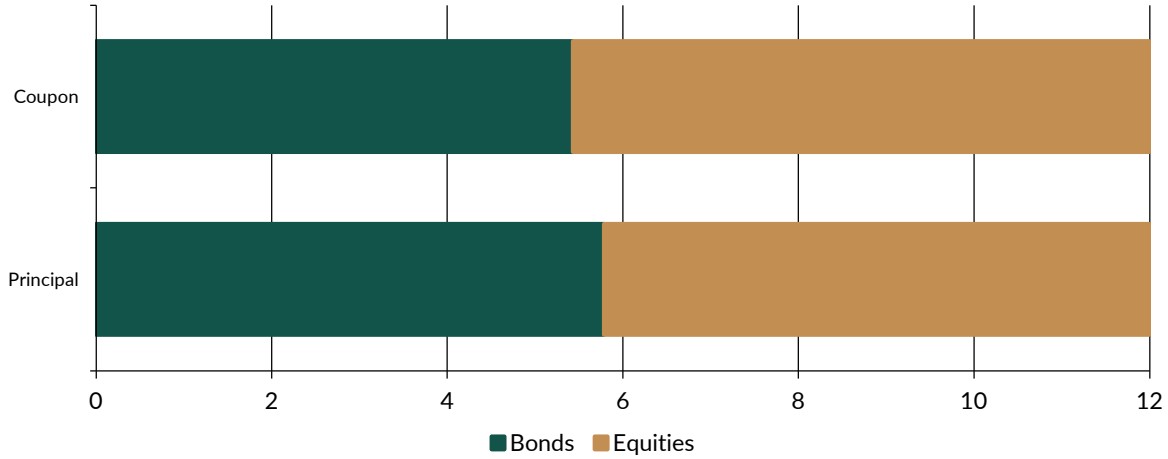


Revised: $2.4m in 2024, $1.2m in 2025, $2.4m in 2027.

We give up growth capital for liquidity

The coupon-funded alternative has more equities but almost no 2025 reserve.

USD millions committed in 2022



Coupon funding adds $354k for equities.

Cash after the 2025 payment

$396k

Principal matching

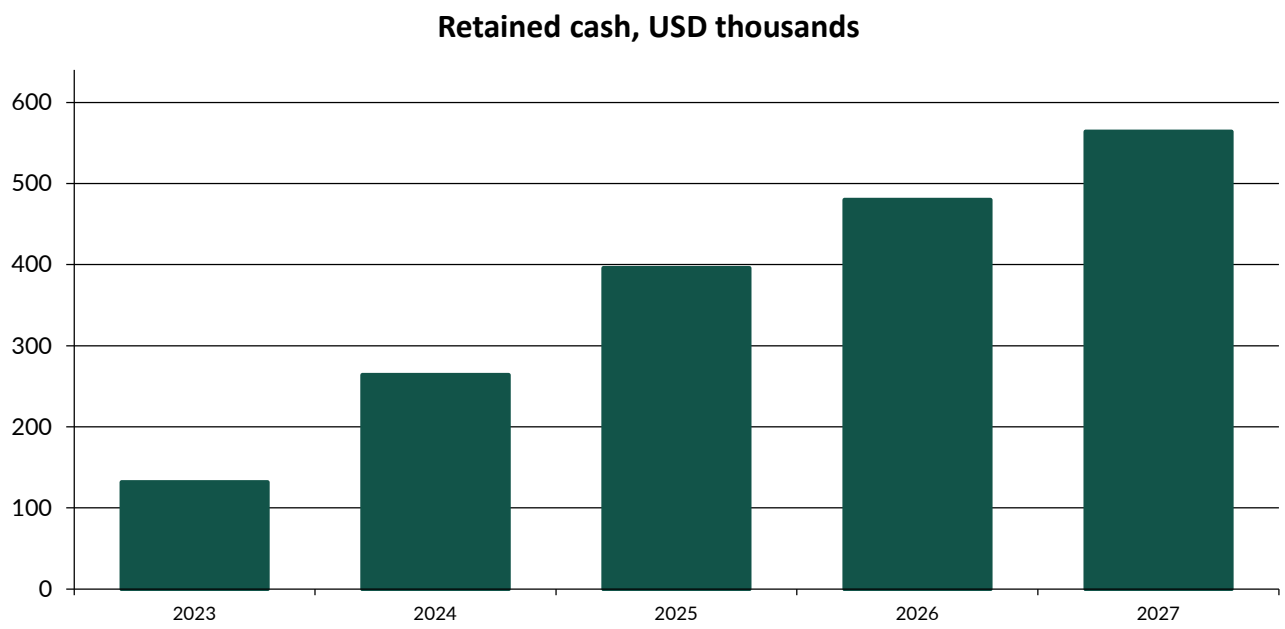
Nearly zero

Coupon funding

Both cover the scheduled payments under our cash-flow assumptions.

The 2025 payment has a cash cushion

Selected bonds mature on time and coupons are retained at zero interest.



2025 principal maturity

$1.20m

Gamma redemption covers the payment

2025 reserve

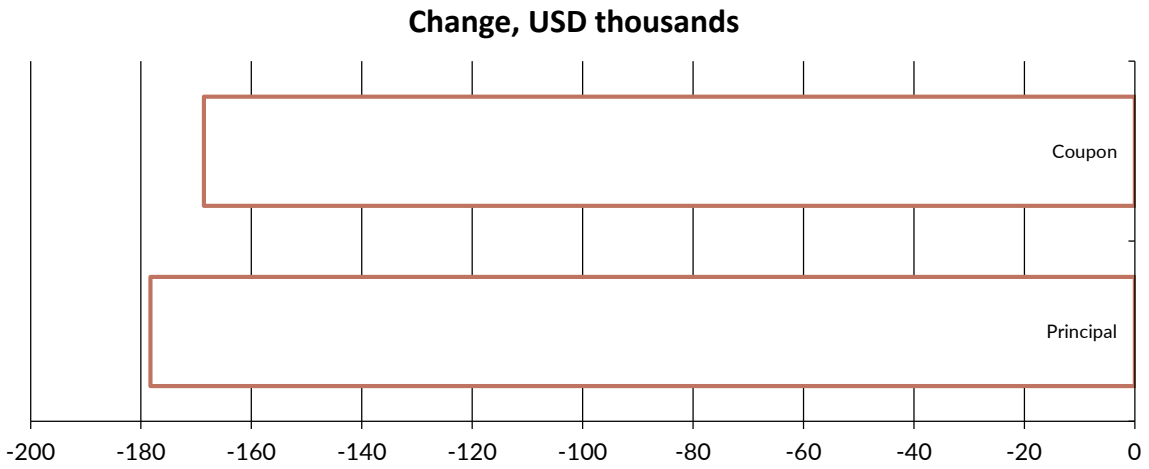
$396,000

The cash balance is after paying the $1.2m commitment.

Bond pricing still responds to yields

Holding bonds to maturity reduces funding risk, not market-price volatility.

Price response to a parallel +100bp shock



A valuation shock is not a missed bond payment.

Case-date present value

$$P = \Sigma\ CFt / (1 + y)^t$$

Rate approximation

$$\Delta P/P \approx -Dmod\ \Delta y + \tfrac{1}{2}\ Convexity\ (\Delta y)^2$$

Bond marks are modeled, not historical market prices.

Where the full $12 million goes

We show the entry fee separately to keep the allocation accurate.



Net equity purchased

$6.210m

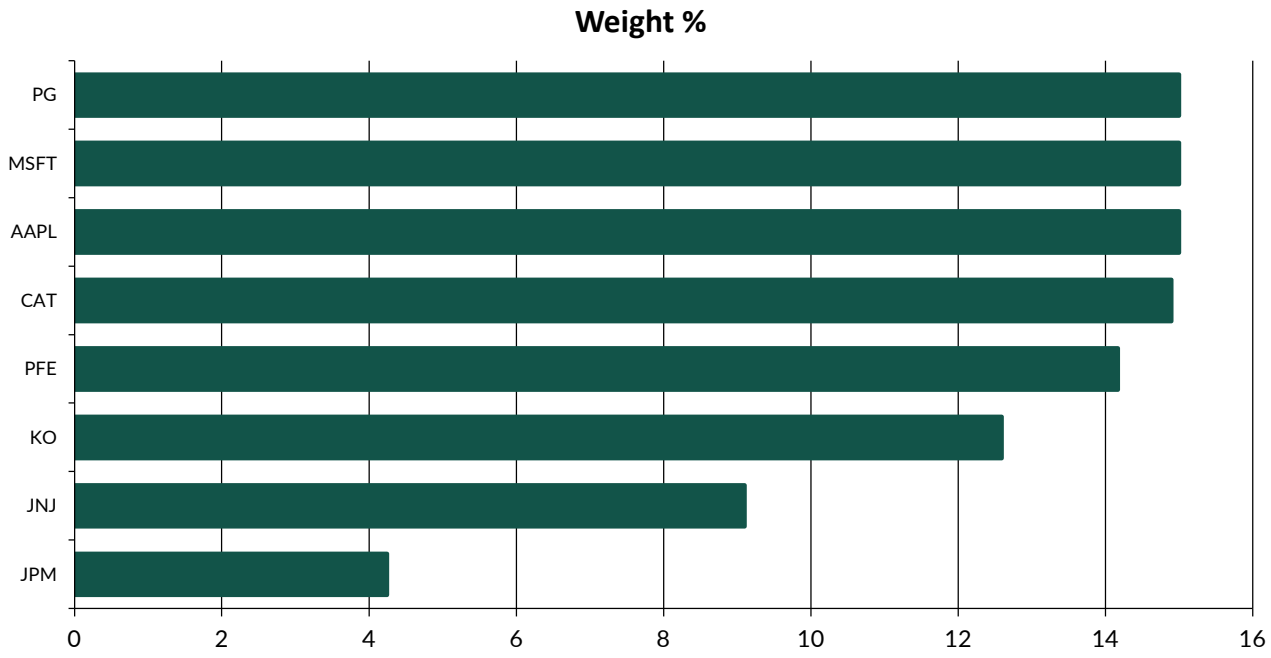
Entry fee

E = (12m − Bond Cost) / 1.001

Ten basis points: about $6,210.

Eight stocks, with strict weight limits

The candidate list spans six sectors; selected stocks span five.



Stock cap

15%

Technology cap

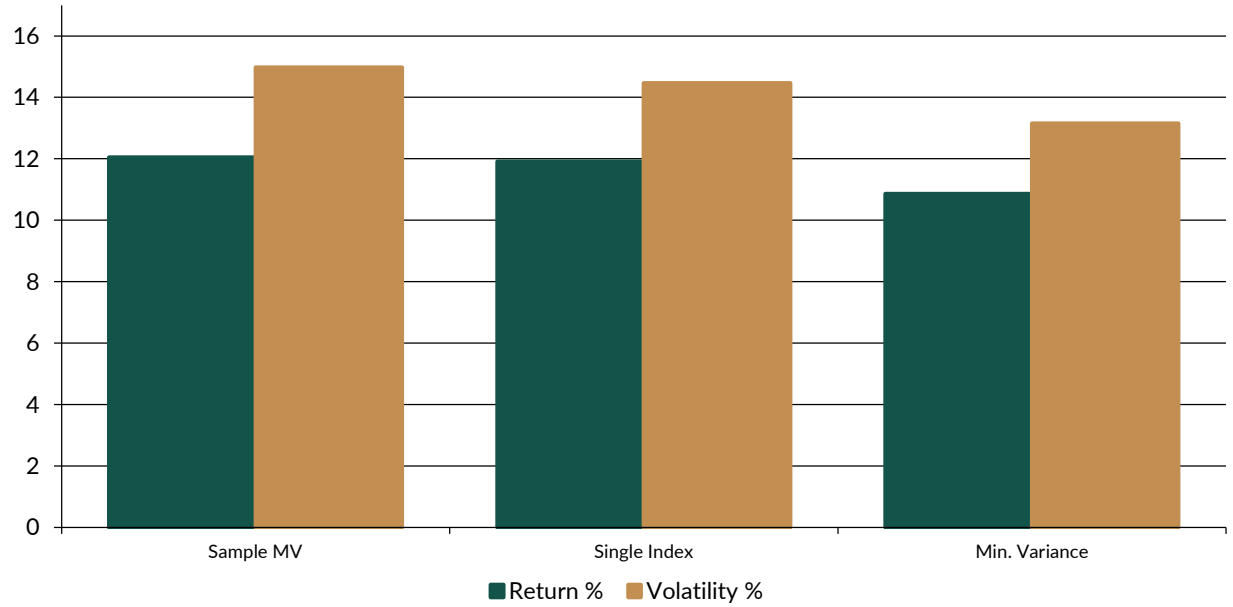
30%

No selected ETFs

Stock concentration still matters.

More expected return means more estimated risk

Model statistics are in sample, not observed 2023–2025 returns.



Sample MV

Full covariance from 96 months

Single Index

Market-factor covariance assumption

Minimum Variance

Reduce estimated risk

What the optimizer actually solves

The forecasts change, but the constraints stay comparable.

Risk-adjusted objective

$$\max \ \mu^T w - (A/2)\, w^T \Sigma w$$

Investment rules

- $\Sigma w = 1;\ w \geq 0$
- Each direct stock $\leq 15\%$
- Direct-stock sector $\leq 30\%$
- Each ETF $\leq 30\%$; combined ETFs $\leq 40\%$

Expected-return blend

$$\mu = 0.25\ \mu_{sample} + 0.75\ \mu_{VTI\ prior}$$

96 months

Jan 2015 – Dec 2022

Foundation utility also scales by the equity share.

Re-optimization delivers a tiny modeled gain

Keeping the frozen shares avoids overreacting to unstable forecasts.

Foundation equity contribution to utility

$$U = F\,\mu_p - (A/2)\,F^2\,\sigma_p^2$$

A = 3, revised Principal Matching

Frozen weights	**0.053328705**
Reoptimized	**0.053328718**

Gain: +0.00000001318 in utility

Reconsider at higher risk aversion

$$A^* \approx 8.94$$

Fixed-policy comparison only

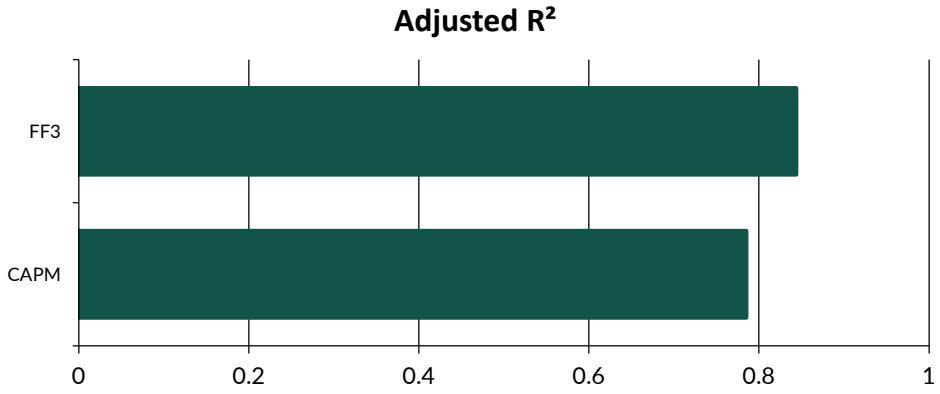
Reassess and re-optimize

Do not automatically buy the old Minimum Variance portfolio.

FF3 fits the historical data better

A positive estimated intercept is not proof of future stock-picking skill.

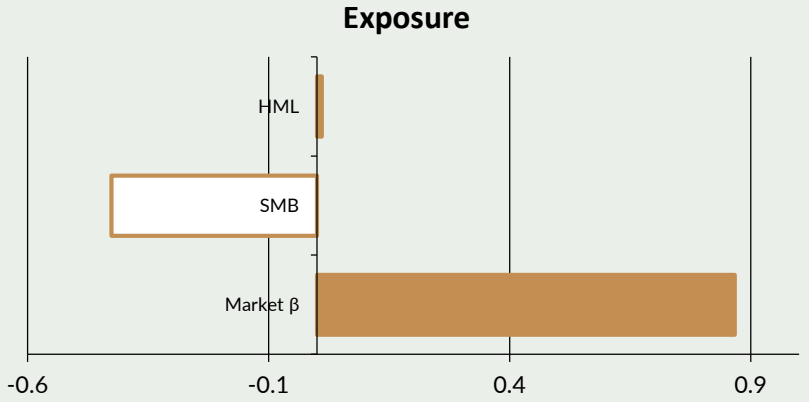
Adjusted R squared



Excess Rp = α + βm excess Rm + ε

FF3 adds SMB and HML.

FF3 factor loadings

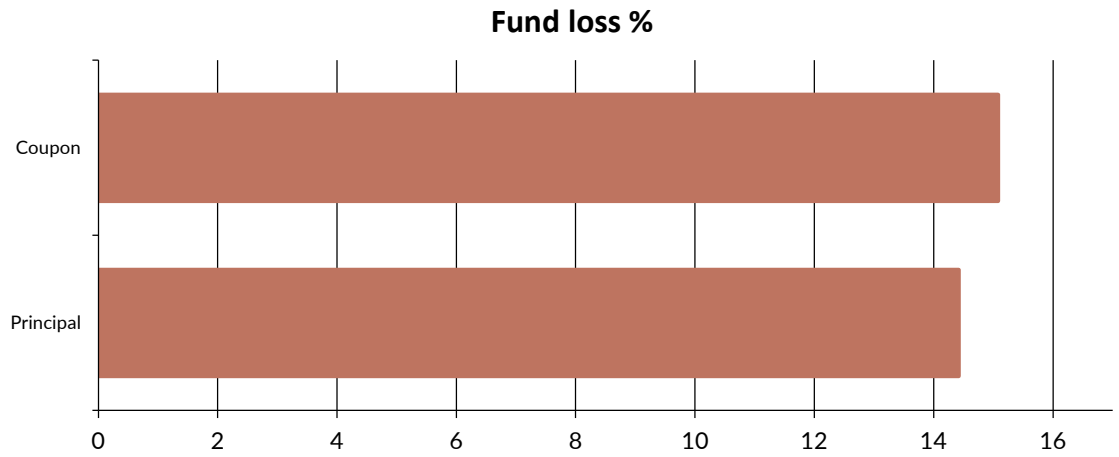


Monthly FF3 α ≈ +0.569% (in sample)

Funding commitments does not eliminate market losses

The 2030 goal also depends on inflation and equity growth.

Combined shock: stocks −25%, yields +100bp



A one-time shock is not a rolling 12-month return.

2030 nominal target

$9.596m

Assuming 2.3% inflation

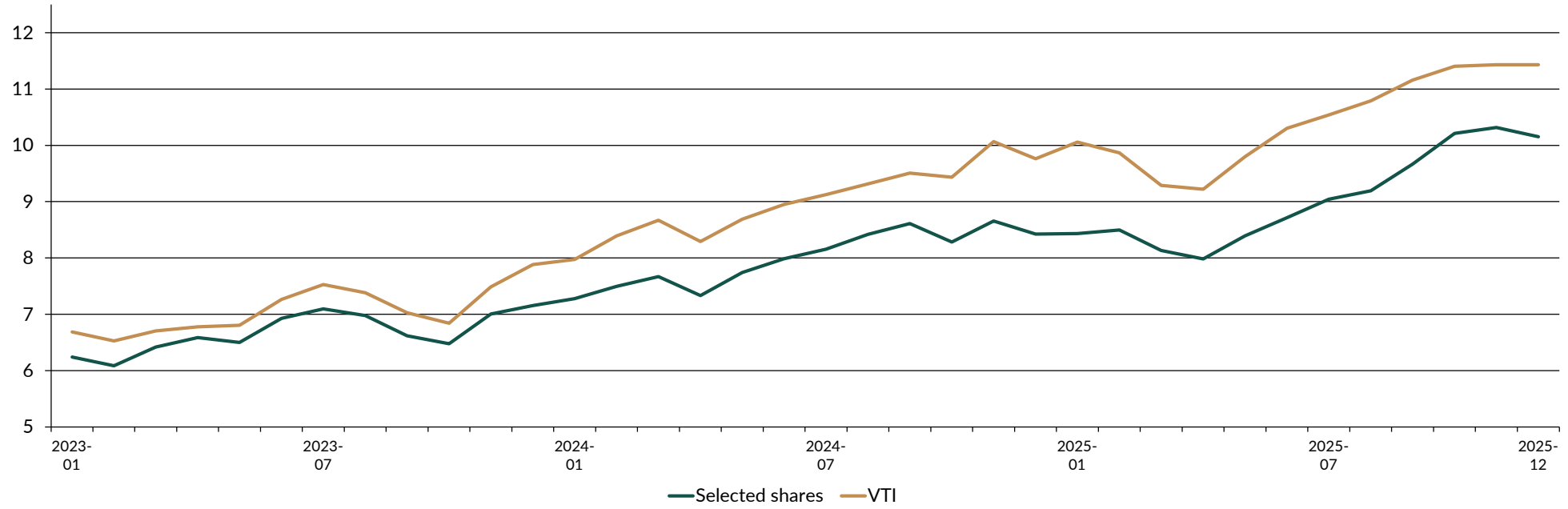
Equity CAGR needed

4.80%

r* = [(Target − Cash) / E]^(1/8) − 1

The selected shares grew, but VTI did better

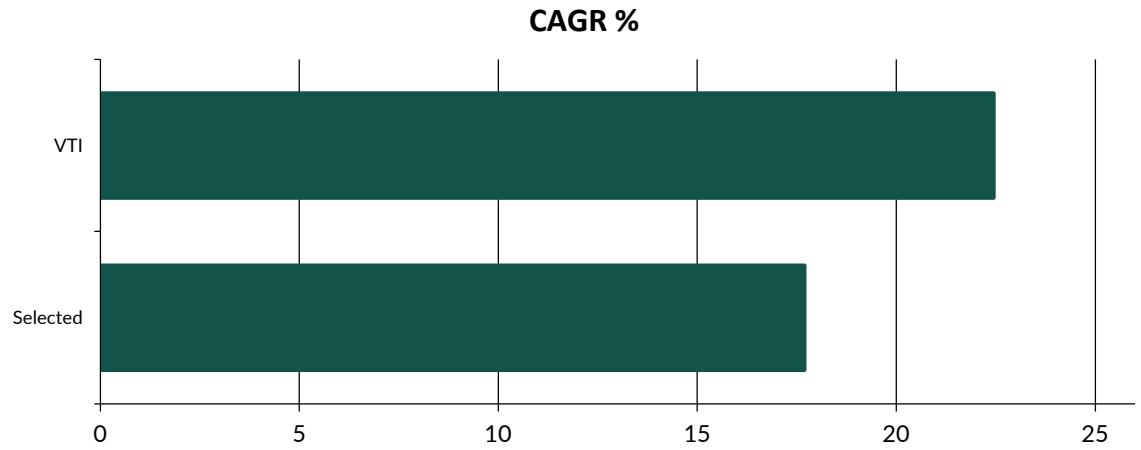
Original $6.229m equity base, not revised $6.210m. Monthly returns, 2023–25.



Selected shares underperformed the VTI benchmark

Positive absolute returns do not prove an equity-selection advantage.

CAGR after entry fee



VTI outperformed by 4.76 percentage points

Observed risk

11.80% vs 12.60%

Annualized volatility

6.91%

Annualized tracking error

The future test did not change our 2022 decision.

Implementation requires monitoring and discipline

Our proposed review rules still require trustee agreement.

2024	2025	2027
$2.4m	$1.2m	$2.4m

Rolling withdrawal-adjusted review levels

12% trailing loss

Warning: review cash and assumptions

15% trailing loss

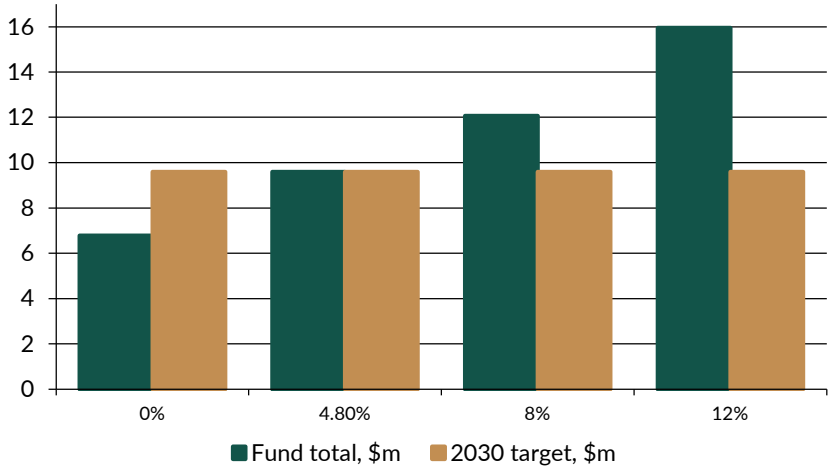
Stronger trustee review

Rolling TWR = Π(1 + withdrawal-adjusted monthly return) − 1

One clear recommendation, with risks disclosed

What we would ask the trustees to approve.

2030 goal under equity scenarios



Return scenarios, not forecasts

2.3% inflation | unspent bond cash at 0%

01 Fund the promises

Principal-match every revised payment.

02 Keep the original equity choice

Retain Sample MV weights without hindsight.

03 Monitor and reconsider

Use cash needs and rolling losses to review.

The 2030 goal is uncertain. Human review and university acceptance remain separate evidence.